Tutorial - 6

① (a) $\hat{\sigma} = \sigma_x \hat{x} + \sigma_y \hat{y} + \sigma_z \hat{z}$

$\hat{n} = \cos\phi \sin\theta \, \hat{x} + \sin\phi \sin\theta \, \hat{y} + \cos\theta \, \hat{z}$

$\therefore \hat{\sigma} \cdot \hat{n} = \cos\phi \sin\theta \, \sigma_x + \sin\phi \sin\theta \, \sigma_y + \cos\theta \, \sigma_z$

$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$; $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$; $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$\therefore \hat{\sigma} \cdot \hat{n} = \begin{pmatrix} \cos\theta & \cos\phi \sin\theta - i \sin\phi \sin\theta \\ \cos\phi \sin\theta + i \sin\phi \sin\theta & -\cos\theta \end{pmatrix}$.

$\hat{\sigma} \cdot \hat{n} = \begin{pmatrix} \cos\theta & \sin\theta \, e^{-i\phi} \\ \sin\theta \, e^{i\phi} & -\cos\theta \end{pmatrix}$. (Ans)

(b) $|\hat{\sigma} \cdot \hat{n} - cI| = 0$. c — Eigenvalues.

$\Rightarrow \begin{vmatrix} \cos\theta - c & \sin\theta \, e^{-i\phi} \\ \sin\theta \, e^{i\phi} & -\cos\theta - c \end{vmatrix} = 0$

$\Rightarrow$ ~~sin2θ~~ $c^2 - \cos^2\theta - \sin^2\theta = 0 \Rightarrow \boxed{c^2 = 1}$ (Ans)

(c) @ $c = 1$:

Let $|\psi\rangle = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}$.

$\therefore \begin{pmatrix} \cos\theta & \sin\theta \, e^{-i\phi} \\ \sin\theta \, e^{i\phi} & -\cos\theta \end{pmatrix} \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}$.

$\Rightarrow \begin{pmatrix} \psi_+ \cos\theta + \psi_- \sin\theta \, e^{-i\phi} \\ \psi_+ \sin\theta \, e^{i\phi} - \psi_- \cos\theta \end{pmatrix} = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}$

Comparing both sides:

$\psi_+ \cos\theta + \psi_- \sin\theta \, e^{-i\phi} = \psi_+$.

$\Rightarrow \psi_+ = \psi_- \sin\theta \, e^{-i\phi} / (1 - \cos\theta)$ — (1)

$$\Rightarrow \frac{\psi_+}{\psi_-} = \frac{\sin\theta\, e^{-i\phi}}{1-\cos\theta} = \frac{\sin\theta\, e^{-i\phi}}{2\sin^2\theta/2}$$

$$\boxed{\frac{\psi_+}{\psi_-} = \frac{\cos\theta/2}{\sin\frac{\theta}{2}\, e^{i\phi}}} \quad \text{— (i)}$$

Normalisat^n:

$$\langle\psi|\psi\rangle = \begin{pmatrix}\psi_+^* & \psi_-^*\end{pmatrix}\begin{pmatrix}\psi_+\\ \psi_-\end{pmatrix} = |\psi_+|^2 + |\psi_-|^2 = 1$$

$$\Rightarrow \boxed{|\psi_+|^2 + |\psi_-|^2 = 1} \quad \text{— (ii)}$$

From (i) & (ii):

$$\psi_+ = \cos\frac{\theta}{2} \quad ; \quad \psi_- = \sin\frac{\theta}{2}\, e^{i\phi}$$

$$\therefore |\psi\rangle = \begin{pmatrix}\cos\frac{\theta}{2}\\ \sin\frac{\theta}{2}\, e^{i\phi}\end{pmatrix} = \cos\frac{\theta}{2}|\uparrow\rangle + \sin\frac{\theta}{2}\, e^{i\phi}|\downarrow\rangle \quad \text{(Ans)}$$

$$|\uparrow\rangle = \begin{pmatrix}1\\0\end{pmatrix} \quad ; \quad |\downarrow\rangle = \begin{pmatrix}0\\1\end{pmatrix}$$

(2) $\hat{L}^2$ Converting to spherical polar coordinates:

$$\psi(r,\theta,\phi) = r\cos\theta\, f(r)$$

$$= \underbrace{r f(r)}_{(i)}\ \underbrace{\cos\theta}_{(ii)}$$

(i) — radial part.

(ii) — angular "

For angular part, we want to substitute it with one of the spherical harmonics as they are eigenf^n of $\hat{L}^2$.

$$\because Y_{10} = \sqrt{\frac{3}{4\pi}}\cos\theta \Rightarrow \cos\theta = \sqrt{\frac{4\pi}{3}}\, Y_{10}.$$

$$\therefore \psi(r,\theta,\phi) = r f(r)\cdot\sqrt{\frac{4\pi}{3}}\cdot Y_{10}.$$

$$= R(r)\, Y_{10} \quad [R(r) \text{ — radial part}]$$

Now, $\hat{L}^2 Y_{10}(\theta,\phi) = \hbar^2 l(l+1)\, Y_{10}(\theta,\phi)$ $\left[Y_{lm} - Y_{10};\ \begin{matrix} l=1 \\ m=0\end{matrix}\right]$.

$$= \hbar^2\cdot 1(1+1)\, R(r)\, Y_{10}(\theta,\phi).$$

$$\Rightarrow \hat{L}^2\psi(r,\theta,\phi) = 2\hbar^2\psi(r,\theta,\phi).$$

$\therefore$ Eigenvalue $= 2\hbar^2 =$ outcome of measurement

Probability $= 100\% = 1$